



Rossmoyne Senior High School

Semester One Examination, 2023

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section One: Calculator-free

MARKING GUIDE

WA student number: In figures

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In words

Your name

Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	53	35
Section Two: Calculator-assumed	12	12	100	101	65
Total					100

Instructions to candidates

- The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
- Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
- You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
- Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
- It is recommended that you do not use pencil, except in diagrams.
- Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
- The Formula sheet is not to be handed in with your Question/Answer booklet.

Markers use only		
Question	Maximum	Mark
1	7	
2	8	
3	8	
4	8	
5	8	
6	7	
7	7	
S1 Total	53	
S1 Wt ($\times 0.6731$)	35%	
S2 Wt	65%	
Total	100%	

Section One: Calculator-free**35% (53 Marks)**

This section has **seven** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

DO NOT WRITE IN THIS AREA AS IT WILL BE CUT OFF

Question 1

(7 marks)

(a) Let $f(x) = 2x^2 - 2x + 1$

- (i) Determine the equation of the line of symmetry for the graph of
- $f(x) = 2x^2 - 2x + 1$
- .

(2 marks)

Solution
$x = -\frac{b}{2a} = -\frac{-(-2)}{2(2)} = \frac{1}{2}$ Equation of line is $x = \frac{1}{2}$.
Specific behaviours
✓ indicates valid approach ✓ correct equation

- (ii) Determine the discriminant of the quadratic
- $2x^2 - 2x + 1 = 0$
- and use it to explain the number of real root(s).

(2 marks)

Solution
$\Delta = b^2 - 4ac = (-2)^2 - 4(2)(1) = -4$ Since $\Delta < 0$ then quadratic has no roots.
Specific behaviours
✓ correct value of Δ ✓ uses Δ to explain number of roots

- (b) Solve the following equations.

(i) $(2x - 1)(x + 7) = 0$.

(1 mark)

Solution
$x = \frac{1}{2}, \quad x = -7$
Specific behaviours
✓ correct solutions

(ii) $x^2 + 2x - 15 = 0$.

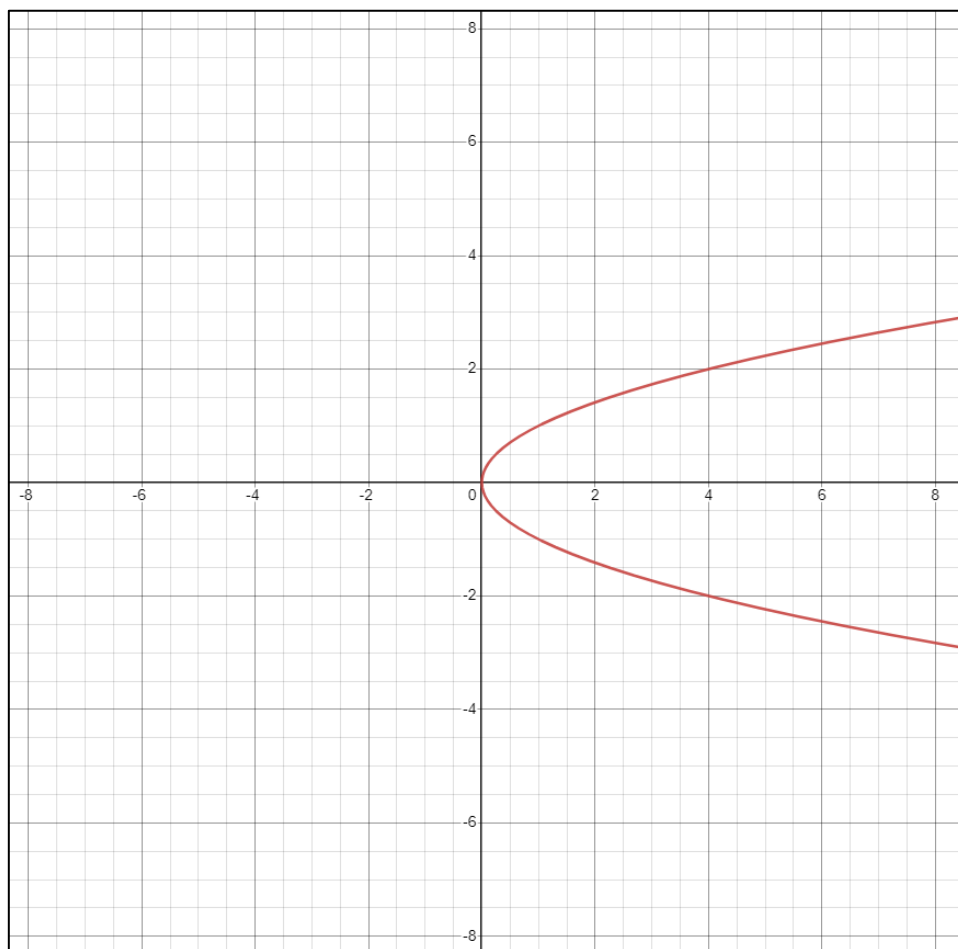
(2 marks)

Solution
$x^2 + 2x - 15 = 0$ $(x - 3)(x + 5) = 0$ $x = 3, \quad x = -5$
Specific behaviours
✓ factorises quadratic ✓ correct solutions

Question 2

(8 marks)

- (a) Sketch the graph of the relationship $x = y^2$ on the axes below, and then use your graph to explain why y is not a function of x . (3 marks)



Solution

See graph (vertical line optional).

A vertical line drawn through graph cuts curve in more than one place and so y is not a function of x (or otherwise indicates one to many relationship).

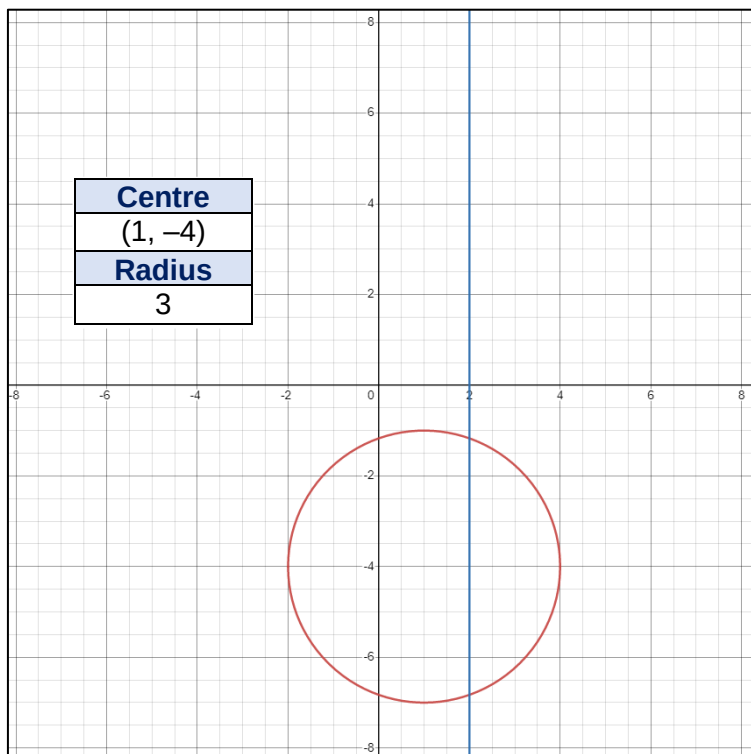
Specific behaviours

- ✓ parabolic shape
- ✓ correct axis of symmetry
- ✓ explains one to many, or uses vertical line test

(b) Another relation has equation $(x - 1)^2 + (y + 4)^2 = 9$.

(i) Sketch the relation on the axes below.

(2 marks)



Solution
See graph
Specific behaviours
✓ circle, centre in correct quadrant
✓ correct circle with scale indicated

(ii) Determine the y-coordinate(s) of the point(s) where the line $x = 2$ cuts the graph of the relation. (3 marks)

Solution
When $x = 2$: $(2 - 1)^2 + (y + 4)^2 = 9$ $(y + 4)^2 = 8$ $y + 4 = \pm\sqrt{8}$ $y = -4 \pm 2\sqrt{2}$
Specific behaviours
✓ substitutes $x = 2$ and obtains quadratic equation ✓ evidence of use of appropriate method ✓ correct y-coordinates

Question 3

(8 marks)

- (a) Determine all solutions of the equation $2 \cos(x - 40^\circ) + \sqrt{3} = 0$ between 0° and 360° .

(3 marks)

Solution
$2 \cos(x - 40^\circ) + \sqrt{3} = 0$ $\cos(x - 40^\circ) = -\frac{\sqrt{3}}{2}$ $x - 40^\circ = 150^\circ, 210^\circ$ $x = 190^\circ, 250^\circ$
Specific behaviours
✓ rearranges ✓ one correct angle for given exact value ✓ both correct solutions

- (b) (i) Using $\cos 2x = \cos(x + x)$, show that $\cos 2x = 2 \cos^2 x - 1$.

(2 marks)

Solution
$\begin{aligned} \cos 2x &= \cos(x + x) \\ &= \cos x \cos x - \sin x \sin x \\ &= \cos^2 x - \sin^2 x \\ &= \cos^2 x - (1 - \cos^2 x) \\ &= 2 \cos^2 x - 1 \end{aligned}$
Specific behaviours
✓ uses angle sum identity and simplifies ✓ uses Pythagorean identity and simplifies

- (ii) Use this identity to obtain an exact value for $\cos 15^\circ$.

(3 marks)

Solution
Let $x = 15^\circ$ $\begin{aligned} \cos(2 \times 15^\circ) &= 2 \cos^2(15^\circ) - 1 \\ \therefore \cos^2(15^\circ) &= \frac{1}{2} \cos(30^\circ) + \frac{1}{2} \\ &= \frac{1}{2} \frac{\sqrt{3}}{2} + \frac{1}{2} \\ &= \frac{\sqrt{3} + 2}{4} \\ \cos(15^\circ) &= \frac{1}{2} \sqrt{\sqrt{3} + 2} \end{aligned}$
Specific behaviours
✓ Sets up equation from part a) i) ✓ obtains correct value ✓ disregards negative solution

Question 4

(8 marks)

(a) Let $f(x) = (x + 2)(8x^2 - 22x + 15)$.(i) Express $f(x)$ in expanded form, without brackets.

(2 marks)

Solution
$ \begin{aligned} f(x) &= (x + 2)(8x^2 - 22x + 15) \\ &= 8x^3 - 22x^2 + 15x + 16x^2 - 44x + 30 \\ &= 8x^3 - 6x^2 - 29x + 30 \end{aligned} $
Specific behaviours
✓ reasonable attempt to expand $f(x)$ ✓ correct expansion

The graph of $y = f(x)$ has roots $x = -2$, $x = \frac{5}{4}$ and $x = \frac{3}{2}$.

(ii) Express $f(x)$ in the form $(x + 2)(ax + b)(cx + d)$, where a, b, c and d are integer constants.

(2 marks)

Solution
$ \begin{aligned} f(x) &= 8(x + 2)\left(x - \frac{5}{4}\right)\left(x - \frac{3}{2}\right) \\ &= (x + 2) \times 4\left(x - \frac{5}{4}\right) \times 2\left(x - \frac{3}{2}\right) \\ &= (x + 2)(4x - 5)(2x - 3) \end{aligned} $
Specific behaviours
✓ uses a and roots to obtain 4 factors ✓ correct factorisation

(b) The graph of $y = 3(x - p)^3 + q$ has a horizontal inflection point at $(-2, 24)$ and y -axis intercept at $(0, k)$, where p, q and k are constants.

(i) the value of k .

(2 marks)

Solution
$(p, q) = (-2, 24)$
Hence $k = 3(0 - (-2))^3 + 24 = 24 + 24 = 48$.
Specific behaviours
✓ indicates values of p and q ✓ correct value

(ii) the root(s) of the graph.

(2 marks)

Solution
$ \begin{aligned} 3(x + 2)^3 + 24 &= 0 \\ (x + 2)^3 &= -8 \\ x + 2 &= -2 \\ x &= -4 \end{aligned} $
Specific behaviours
✓ forms equation and simplifies ✓ correct root

See next page

Question 5

(8 marks)

- (a) State the exact value of $\tan\left(\frac{\pi}{6}\right)$.

(1 mark)

Solution
$\tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$
Specific behaviours
✓ correct value

- (b) Express in simplest form the angle

- (i) 16° in radians.

(1 mark)

Solution
$16^\circ = \frac{16 \times \pi}{180} = \frac{4\pi}{45}$
Specific behaviours
✓ correct value

- (ii) $\frac{3\pi}{10}$ in degrees.

(1 mark)

Solution
$\frac{3\pi}{10} \times \frac{180}{\pi} = 54^\circ$
Specific behaviours
✓ correct value

- (c) Given that angle $\theta \in \left\{0^\circ, \frac{7\pi}{8}, 270^\circ, \pi, -100^\circ, \frac{\pi}{2}, 21^\circ, -\frac{\pi}{20}\right\}$, state which, if any, of the angles are such that

- (i) $\cos \theta < 0$.

(1 mark)

Solution
$\theta = \frac{7\pi}{8}, \pi, -100^\circ$
Specific behaviours
✓ correct set of values

- (ii) $\tan \theta$ is undefined.

(1 mark)

Solution
$\theta = 270^\circ, \frac{\pi}{2}$
Specific behaviours
✓ correct set of values

- (iii) $\tan \theta < 0$ and $\sin \theta > 0$.

(1 mark)

Solution
$\theta = \frac{7\pi}{8}$
Specific behaviours
✓ correct value

- (d) Given that $\sin \alpha = \frac{3}{5}$ and $\cos \beta = \frac{1}{5}$, determine the value of $\sin(-\alpha) + \cos(\beta + \pi)$.

(2 marks)

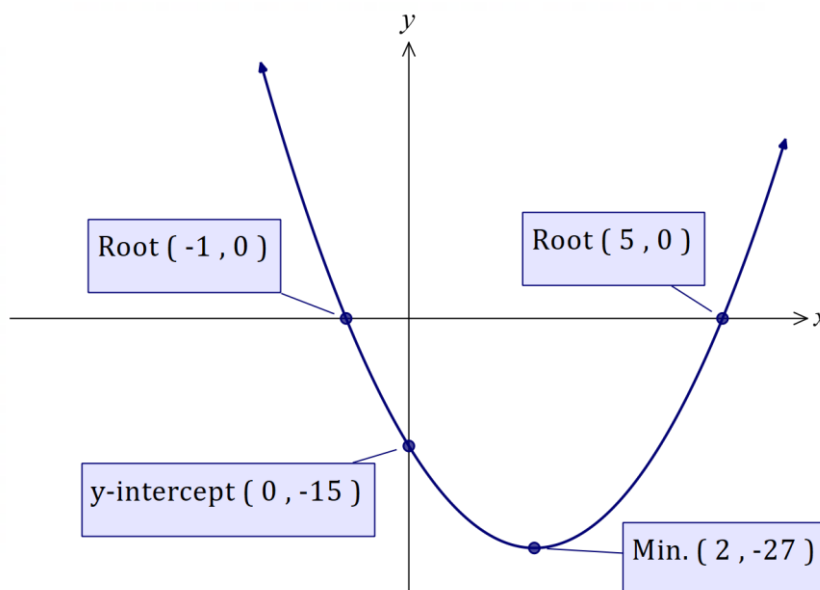
Solution
Using symmetry of unit circle: $\sin(-\alpha) + \cos(\beta + \pi) = -\sin(\alpha) - \cos(\beta) = \left(-\frac{3}{5}\right) + \left(-\frac{1}{5}\right) = -\frac{4}{5}$
Specific behaviours
✓ correct value for $\sin(-\alpha)$ ✓ correct value for $\cos(\beta + \pi)$ and sum

Question 6

(7 marks)

- (a) Sketch the graph of
- $y = 3(x - 2)^2 - 27$
- below.

(4 marks)

**Solution**TP: $(2, -27)$. Intercept: $x = 0 \rightarrow y = 12 - 27 = -15$.Roots: $(x - 2)^2 = 27 \div 3 = 9 \rightarrow x - 2 = \pm\sqrt{9} \rightarrow x = 2 \pm 3$.

See graph – plots 4 key points and uses to draw smooth parabolic curve.

Specific behaviours

✓ turning point ✓ y-intercept ✓ roots ✓ smooth parabolic curve

- (b) The graph of
- $y = ax^2 + bx + c$
- has roots
- $x = -10$
- and
- $x = 5$
- and passes through the point
- $(-5, 20)$
- . Determine the value of
- a
- , the value of
- b
- and the value of
- c
- . (3 marks)

Solution

$$y = a(x + 10)(x - 5)$$

$$(-5, 20) \Rightarrow 20 = a(5)(-10) \Rightarrow a = -\frac{20}{50} = -\frac{2}{5}$$

$$\begin{aligned} y &= -\frac{2}{5}(x + 10)(x - 5) \\ &= -\frac{2}{5}(x^2 + 5x - 50) \\ &= -\frac{2}{5}x^2 - 2x + 20 \end{aligned}$$

$$\text{Hence } a = -\frac{2}{5}, b = -2 \text{ and } c = 20.$$

Specific behaviours

- ✓ uses roots to write in factored form
- ✓ uses point to determine a
- ✓ expands quadratic and states coefficients

Question 7

(7 marks)

Consider the following expansion of the binomial $(x + 1)^{11}$:

$$x^{11} + 11x^{10} + kx^9 + 165x^8 + 330x^7 + 462x^6 + 462x^5 + 330x^4 + 165x^3 + 55x^2 + 11x + 1.$$

- (a) State the degree of the above polynomial and the value of the constant k . (2 marks)

Solution
Degree is 11 and $k = 55$.
Specific behaviours
✓ correct degree
✓ correct value of k

- (b) There are m ways to choose 3 cats and n ways to choose 8 cats from a choice of 11 different cats. Use coefficients from the expansion to explain whether m is greater than, equal to, or less than n . (2 marks)

Solution
$m = \binom{11}{3} = 165$ and $n = \binom{11}{8} = 165$. Hence $m = n$.
Specific behaviours
✓ one correct value
✓ two correct values and correct response

The first three terms in the expansion of $(x + 2)^{10}$ are $x^{10} + 20x^9 + ax^8$.

- (c) Determine the value of a and the coefficient of the x^7 term in the expansion. (3 marks)

Solution
$ax^8 = \binom{10}{8} x^8 (2)^2, \quad \binom{10}{8} = \frac{10!}{8!2!} = \frac{10 \times 9}{2} = 45$ $ax^8 = 45x^8 \times 4 \Rightarrow a = 180$ Next term is $\binom{10}{7} x^7 (2)^3$. $\binom{10}{7} = \frac{10!}{7!3!} = \frac{10 \times 9 \times 8}{3 \times 2} = 10 \times 3 \times 4 = 120$ Hence coefficient of x^7 term is $\binom{10}{7} (2)^3 = 120 \times 8 = 960$. Note that coefficients can also be derived using properties of Pascal's triangle: $\binom{n}{r} = \binom{n+1}{r+1} - \binom{n}{r+1}$ so that $\binom{10}{8} = \binom{11}{9} - \binom{10}{9} = 55 - 10 = 45$ and so on.
Specific behaviours
✓ correct value of $^{10}C_8$
✓ correct value of a
✓ correct next term

Supplementary page

Question number: _____

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